

OJT Project Design Template

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Year & Section:2025, semester 3

Project Title (as assigned): Course Enrollment API

Project Type: Application development

Stack / Framework:FastAPI, SQLAlchemy

1. Problem Understanding

1.1 What is the problem statement in your own words?

I need to build a system to store students, courses, and grades in a database. Teachers need a way to manage which students are in which courses and what grades they got.

1.2 Why does this problem exist or matter?

Right now there is no digital way to track course enrollments and calculate grades. This will help schools manage student information easily.

1.3 Key inputs and expected outputs:

Inputs	Process	Expected Outputs
Student name, email	Save to database	Student created
Course name,code	Save to database	Course created
Student enrolls in course	Link student and course	Enrollment record

Grade marks	Calculate grade	grade number
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2. Functional Scope

2.1 What are the core features you plan to build (must-haves)?

- Add, view, update, and delete students
- Add, view, update, and delete courses
- Enroll a student in a course
- Add grades and calculate final grade
- Check that inputs are correct (like valid email format)

2.2 What stretch goals could you attempt if time permits?

- Export grade report as PDF file (only if time is left)

2.3 Which libraries or tools will you use?

Tools i will be using->

1. FastAPI - To build the API (required by assignment)
2. SQLAlchemy: To work with database (required by assignment)
3. Postman: To test my 6 API endpoints

3. System & Design Thinking

3.1 Sketch or describe your app flow / pipeline:

User sends request using Postman → FastAPI receives request → Check if data is correct → SQLAlchemy saves/gets from database → Send result back

3.2 What data structures or algorithms are central to this project?

- 4 simple database tables: Students, Courses, Enrollments, Grades
- Links between tables: One student can take many courses
- Grade calculation: Simple math formula to calculate final grade

Simple table design:

- Students: id, name, email
- Courses: id, course_name, course_code, credits
- Enrollments: id, student_id, course_id, enrollment_date
- Grades: id, enrollment_id, marks, final_grade

3.3 How will you test correctness or performance?

- Use Postman to test all features
- Create 6 saved requests in Postman (add student, get student, add course, get course, enroll student, add grade)
- Take screenshots showing tests working
- FastAPI also creates automatic test page at /docs

4. Timeline & Milestones (4 Weeks)

Week	Planned Deliverables	Mentor Checkpoint
W1	- Watch FastAPI beginner tutorial - Watch SQLAlchemy beginner tutorial - Set up project folders - Create 4 database tables (Students, Courses, Enrollments, Grades)	☐
W2	- Build add student feature - Build view all students feature - Build update and delete student - Build same 4 features for courses - Test using Postman	☐
W3	- Build enroll student in course feature - Build add grade feature - Build calculate final grade feature - Test all features using Postman	☐
W4	- Create 6 saved test requests in Postman - Take screenshots of all tests passing	☐

	- Write simple README file - Record 2-3 minute demo video - Submit everything	
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5. Risks & Dependencies

5.1 What's the hardest part technically for you right now?

- Never used FastAPI before - need to learn from tutorials
- Never used SQLAlchemy before - need to learn from tutorials
- Understanding how to link tables together (student to course)
- Making sure grade calculation is correct

5.2 What dependencies or help do you need from mentors?

- Help installing FastAPI and SQLAlchemy
 - Check if my 4 tables are designed correctly (Week 1)
 - Help if I get errors while coding
 - Verify grade calculation is working correctly
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6. Evaluation Readiness

6.1 How will you prove that your project "works"?

- Demo video: 2-3 minute recording showing all features
- Postman screenshots: Show 6 test requests working
- API documentation: FastAPI creates this automatically at [/docs](#)
- README file: Setup instructions
- GitHub: Code uploaded

6.2 What success metric or goal will you aim for?

- Grade calculation correct (main evaluation metric as per assignment)
- All CRUD operations work (add, view, update, delete)
- 6 test requests in Postman all pass
- Automatic API documentation works

7. Responsibilities

7.1 Responsibilities

Task	Student Name	Mentor Notes
Learn FastAPI and SQLAlchemy tutorials (Week 1)	Arpit tripathi	
Create 4 database tables (Week 1)		
Build student and course CRUD (Week 2)		
Build enrollment and grades (Week 3)		
Test with Postman and make video (Week 4)		

Signatures (Students): Arpit tripathi

Mentor Approval:

Date: